SESSION 9 - JOINS - Part 1

Task 1:Create two tables in your database: 'restaurants' (id, name, city) and 'dishes' (id, restaurant_id, dish_name, price). Insert at least 3 restaurants and 2-3 dishes for each restaurant.
Query / Answer:

```
create table restaurants1(

        res_id int AUTO_INCREMENT primary key ,
   res_name varchar(50) not null,
   res_city varchar(50) not null

);

create table dishes1(
   dish_id int auto_increment primary key,
   restaurant_id int,
   dish_name varchar(50) ,
   price int,

   foreign key (restaurant_id) references restaurants1(res_id));

   INSERT INTO restaurants1 (res_name, res_city) VALUES
('Barbeque Nation', 'Ahmedabad'),
('Dominos Pizza', 'Surat'),
('The Grand Thakar', 'Ahmedabad'),
('KFC', 'Vadodara'),
('Pizza Hut', 'Rajkot');

INSERT INTO dishes1 (restaurant_id, dish_name, price) VALUES
(1, 'Paneer Tikka', 350),
(1, 'Veg Biryani', 280),
(2, 'Margherita Pizza', 249),
(2, 'Farmhouse Pizza', 399),
(3, 'Gujarati Thali', 450),
(3, 'Dal Khichdi', 220),
(4, 'Chicken Bucket', 699),
(4, 'Zinger Burger', 199),
(5, 'Veg Supreme Pizza', 449),
(5, 'Garlic Bread', 149);
```

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Image / Screenshot:

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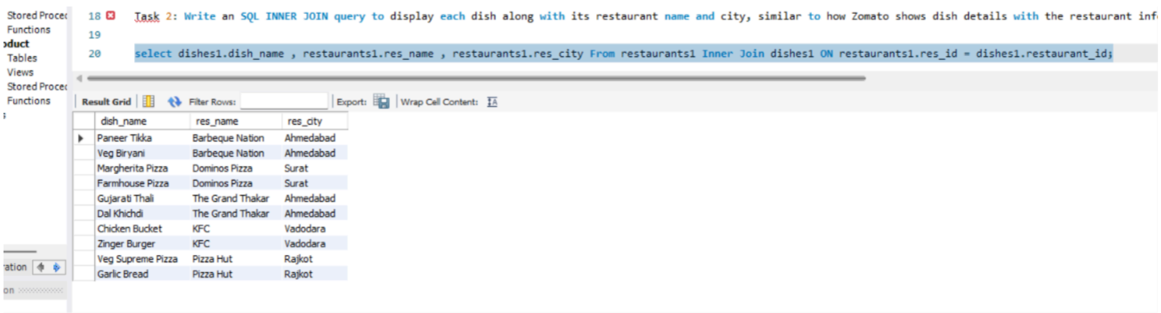
Task 2: Write an SQL INNER JOIN query to display each dish along with its restaurant name and city, similar to how Zomato shows dish details with the restaurant info.
Query / Answer: select dishes1.dish_name , restaurants1.res_name , restaurants1.res_city From restaurants1 Inner Join dishes1 ON restaurants1.res_id = dishes1.restaurant_id;
Image / Screenshot:



dish_name	res_name	res_city
Paneer Tikka	Barbeque Nation	Ahmedabad
Veg Biryani	Barbeque Nation	Ahmedabad
Margherita Pizza	Dominos Pizza	Surat
Farmhouse Pizza	Dominos Pizza	Surat
Gujarati Thali	The Grand Thakar	Ahmedabad
Dal Khichdi	The Grand Thakar	Ahmedabad
Chicken Bucket	KFC	Vadodara
Zinger Burger	KFC	Vadodara
Veg Supreme Pizza	Pizza Hut	Rajkot
Garlic Bread	Pizza Hut	Rajkot

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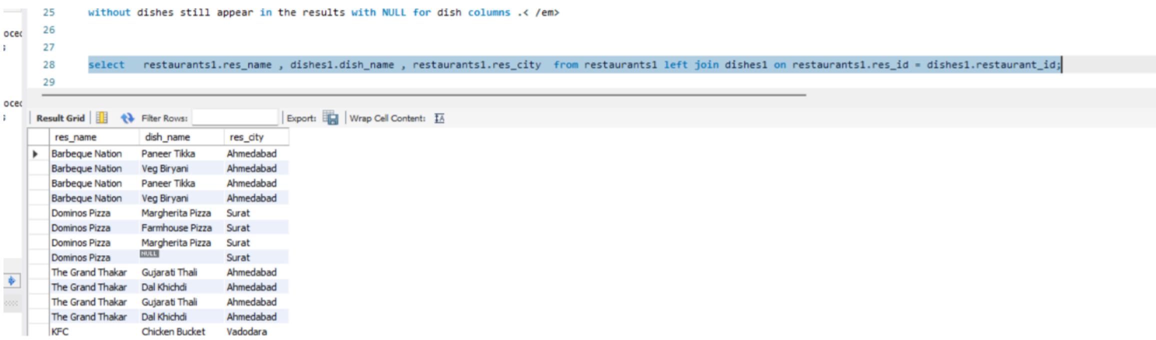
Task 3: Write an SQL LEFT JOIN query to list all restaurants and their dishes, showing restaurants even if they currently have no dishes on the menu .< br> <em><strong>Hint :< /strong> Use LEFT JOIN so restaurants without dishes still appear in the results with NULL for dish columns .< /em>
Query / Answer: select   restaurants1.res_name , dishes1.dish_name , restaurants1.res_city from restaurants1 left join dishes1 on restaurants1.res_id = dishes1.restaurant_id;
Image / Screenshot:



res_name	dish_name	res_city
Barbeque Nation	Paneer Tikka	Ahmedabad
Barbeque Nation	Veg Biryani	Ahmedabad
Barbeque Nation	Paneer Tikka	Ahmedabad
Barbeque Nation	Veg Biryani	Ahmedabad
Dominos Pizza	Margherita Pizza	Surat
Dominos Pizza	Farmhouse Pizza	Surat
Dominos Pizza	Margherita Pizza	Surat
Dominos Pizza	NULL	Surat
The Grand Thakar	Gujarati Thali	Ahmedabad
The Grand Thakar	Dal Khichdi	Ahmedabad
The Grand Thakar	Gujarati Thali	Ahmedabad
The Grand Thakar	Dal Khichdi	Ahmedabad
KFC	Chicken Bucket	Vadodara

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Task 4: Write an SQL RIGHT JOIN query to display all dishes and their restaurant names, including any dishes that might not be linked to a restaurant (simulate a data error where a dish has a restaurant_id that doesn't match any restaurant).
Query / Answer: select dishes1.dish_name , restaurants1.res_name , restaurants1.res_city from restaurants1 right join dishes1 on restaurants1.res_id = dishes1.restaurant_id; SET FOREIGN_KEY_CHECKS = 0; INSERT INTO dishes1 (restaurant_id, dish_name, price) VALUES (99, 'Mystery Dish', 500); SET FOREIGN_KEY_CHECKS = 1;
Image / Screenshot: 

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Task 5: Given this scenario: You want to show a list of all playlists and the songs inside them, like Spotify. Explain which JOIN type (INNER, LEFT, or RIGHT) you would use to show all playlists, even if some are empty, and write the SQL query for it.
Query / Answer: Left join because we want the playlist also which were created but songs were not added yet
Image / Screenshot: